MCQS

ZOO403 midterm preparation file

ZB

بسم الله الرحمن الرحيم

In the name of God , Most Gracious, Most merciful

ZOO403

MCQS

1. Which organelle is responsible for energy production in cells?

a) Nucleus

b) Mitochondria

c) Ribosome

d) Golgi apparatus

Answer: b) Mitochondria

2. What is the basic structural and functional unit of life?

a) Atom

b) Cell

c) Tissue

d) Organ

Answer: b) Cell

3. The genetic material in prokaryotic cells is located in:

a) Nucleus

b) Nucleoid

c) Mitochondria

d) Lysosomes

Answer: b) Nucleoid

4. Which of the following is a eukaryotic organism?

a) Bacteria

b) Fungi

c) Virus

d) Cyanobacteria

Answer: b) Fungi

MCQS

Midterm preparation file

ZB

5. Which phase of mitosis involves the separation of chromatids?

a) Prophase

b) Metaphase

c) Anaphase

d) Telophase

Answer: c) Anaphase

6. What is the site of protein synthesis in a cell?

a) Ribosome

b) Lysosome

c) Golgi apparatus

d) Endoplasmic reticulum

Answer: a) Ribosome

7. Which macromolecule is the main component of cell membranes?

a) Proteins

b) Lipids

c) Carbohydrates

d) Nucleic acids

Answer: b) Lipids

8. In which organelle does photosynthesis occur?

a) Mitochondria

b) Chloroplast

c) Ribosome

d) Endoplasmic reticulum

Answer: b) Chloroplast

9. Which of the following is NOT a type of RNA?

a) mRNA

b) tRNA

c) cRNA

d) rRNA

Answer: c) cRNA

10. Which enzyme is responsible for DNA replication?

a) DNA polymerase

b) RNA polymerase

c) Helicase

d) Ligase

Answer: a) DNA polymerase

11. What is the function of lysosomes?

a) Protein synthesis

b) Energy production

c) Waste digestion

d) Transport of molecules

Answer: c) Waste digestion

12. Which bond joins amino acids in a protein?

a) Glycosidic bond

b) Peptide bond

c) Hydrogen bond

d) Ionic bond

Answer: b) Peptide bond

13. The powerhouse of the cell is:

a) Chloroplast

b) Nucleus

c) Mitochondria

d) Golgi apparatus

Answer: c) Mitochondria

14. Which nitrogenous base is NOT found in RNA?

a) Adenine

b) Guanine

c) Thymine

d) Uracil

Answer: c) Thymine

15. Which type of cell lacks a nucleus?

a) Eukaryotic cell

b) Prokaryotic cell

c) Animal cell

d) Plant cell

Answer: b) Prokaryotic cell

16. The process by which cells engulf solid particles is called:

a) Phagocytosis

b) Pinocytosis

c) Exocytosis

d) Diffusion

Answer: a) Phagocytosis

17. Who proposed the cell theory?

a) Watson and Crick

b) Schleiden and Schwann

c) Mendel

d) Pasteur

Answer: b) Schleiden and Schwann

18. Which phase in the cell cycle involves DNA replication?

a) G1 phase

b) S phase

c) G2 phase

d) M phase

Answer: b) S phase

19. What is the semi-fluid matrix inside chloroplasts called?

a) Stroma

b) Thylakoid

c) Grana

d) Cytoplasm

Answer: a) Stroma

20. The site of lipid synthesis in cells is:

a) Rough ER

b) Smooth ER

c) Golgi apparatus

d) Ribosome

Answer: b) Smooth ER

21. Which type of transport requires energy?

a) Passive transport

b) Osmosis

c) Active transport

d) Diffusion

Answer: c) Active transport

22. The organelle responsible for packaging and secretion is:

a) Lysosome

b) Ribosome

c) Golgi apparatus

d) Nucleus

Answer: c) Golgi apparatus

23. A structure present in plant cells but absent in animal cells is:

a) Mitochondria

b) Cell wall

c) Ribosome

d) Cytoplasm

Answer: b) Cell wall

24. The sugar found in RNA is:

a) Deoxyribose

b) Ribose

c) Glucose

d) Sucrose

Answer: b) Ribose

Midterm preparation file

25. Which process converts glucose into pyruvate?

a) Glycolysis

b) Krebs cycle

c) Electron transport chain

d) Fermentation

Answer: a) Glycolysis

26. Which component of the cytoskeleton is involved in cell division?

a) Microtubules

b) Actin filaments

c) Intermediate filaments

d) Centrosome

Answer: a) Microtubules

27. What is the main pigment responsible for photosynthesis?

a) Carotenoids

b) Chlorophyll

c) Anthocyanins

d) Xanthophylls

Answer: b) Chlorophyll

28. The double-layered structure of the cell membrane is due to:

a) Proteins

b) Phospholipids

c) Cholesterol

d) Carbohydrates

Answer: b) Phospholipids

Midterm preparation file

29. Which organelle contains its own DNA?

a) Golgi apparatus

b) Ribosome

c) Mitochondria

d) Lysosome

Answer: c) Mitochondria

30. Which process occurs in the cytoplasm?

a) Photosynthesis

b) Glycolysis

c) Krebs cycle

d) Electron transport chain

Answer: b) Glycolysis

31. The organelle involved in detoxification of drugs is:

a) Rough ER

b) Smooth ER

c) Golgi apparatus

d) Peroxisome

Answer: b) Smooth ER

32. The tails of phospholipids in the cell membrane are:

a) Hydrophobic

b) Hydrophilic

c) Ionic

d) Amphipathic

Answer: a) Hydrophobic

Midterm preparation file

33. What are the protein filaments in the cytoskeleton primarily made of?
a) Actin
b) Tubulin
c) Keratin
d) All of the above
Answer: d) All of the above

34. Which of the following is a characteristic of active transport?
a) Requires energy
b) Moves molecules against the concentration gradient
c) Involves carrier proteins
d) All of the above
Answer: d) All of the above

35. Which organelle is involved in the production of ATP?
a) Golgi apparatus
b) Mitochondria
c) Ribosomes
d) Lysosomes
Answer: b) Mitochondria

36. Which type of RNA carries amino acids to the ribosome?
a) mRNA
b) tRNA
c) rRNA
d) snRNA
Answer: b) tRNA

Midterm preparation file

37. What is the role of the Golgi apparatus in the cell?

a) Energy production

b) Protein modification and packaging

c) Genetic material storage

d) Waste digestion

Answer: b) Protein modification and packaging

38. What is the fluid mosaic model?

a) A theory explaining DNA replication

b) A description of the cell membrane structure

c) A mechanism of protein synthesis

d) A process of energy production

Answer: b) A description of the cell membrane structure

39. What is the purpose of centrioles in cell division?

a) DNA replication

b) Chromosome alignment

c) Formation of spindle fibers

d) Cytoplasmic division

Answer: c) Formation of spindle fibers

40. Which of the following is an example of passive transport?

a) Osmosis

b) Diffusion

c) Facilitated diffusion

d) All of the above

Answer: d) All of the above

ZB

MCQS

Midterm preparation file

41. In cellular respiration, the electron transport chain occurs in the:

a) Cytoplasm

b) Mitochondrial matrix

c) Inner mitochondrial membrane

d) Nucleus

Answer: c) Inner mitochondrial membrane

42. Which macromolecule forms enzymes?

a) Lipids

b) Proteins

c) Carbohydrates

d) Nucleic acids

Answer: b) Proteins

43. The thickest component of the cytoskeleton is:

a) Microtubules

b) Actin filaments

c) Intermediate filaments

d) Flagella

Answer: a) Microtubules

44. Which molecule is a primary carrier of energy in cells?

a) ATP

b) NADPH

c) FADH2

d) Glucose

Answer: a) ATP

45. What is the main function of the nucleolus?

a) DNA replication

b) RNA synthesis

c) Ribosome production

d) Protein synthesis

Answer: c) Ribosome production

46. What is the primary role of chlorophyll in photosynthesis?

a) Transport of water

b) Absorption of light energy

c) Synthesis of glucose

d) Release of oxygen

Answer: b) Absorption of light energy

ZB +92 300 5427579

WhatsApp:

https://whatsapp.com/channel/0029VaODheTDp2Q8YVcoON2p

Second WhatsApp channel

channel on WhatsApp:

https://whatsapp.com/channel/0029VafvOcy90x2tPbUaX02I

YouTube:

https://youtu.be/-AMj4iNz0UE?si=hg29pJZ1kujzg57J

Phone no : +92 300 5427579

Phone no: +92 330 3869586

Email: asktozb@gmail.com

Passing criteria of this course

Letter Grade	Grade Points	Equivalent Percentage
A+	4.00	90-100
A	4.00	85-89
A-	3.66 - 3.99	80-84
B+	3.33 - 3.65	75-79
B	3.00 - 3.32	71-74
B-	2.66 - 2.99	68-70
C	2.00 - 2.65	61-67
D	1.00 - 1.99	50-60
F	0.00	00-49

Passing Criteria for a Course:

The passing criteria is defined in such a way that it ensures the student shall be consistent in his studies throughout the semester.
Therefore, for passing a course/subject, student shall fulfill the following:
a) Secure minimum 20% score in Formative Assessments/Mid terms
b) Secure minimum 20% score in Final Term Examinations.
c) Secure at least 40% marks in aggregate while fulfilling the above requirements

Course Selection and Credit Hours

Q: How do I select courses when the course selection is open?
A: Follow these steps:
1. Check the credit hours allowed by your university (e.g., 21, 18, or 15 credit hours).
2. Each subject typically has 3 credit hours.
3. Divide the total allowed credit hours by 3 to determine how many subjects you can select.
Examples:
21 credit hours = 7 su18 credit hours = 6 subjects
15 credit hours = 5 subjects
Remember, the university may allow different credit hours for each student, so check your specific allowance.
By following these steps, you can make informed decisions during course selection and manage your credit hours effectively.

ZB

Advice

Hmaesha relax ho ke parha kryn panic mat hua kryn apky parents ky bohat khawab hoty hn wo pura krny ki puri koshish kea kryn kbi b ksi pe depend na kea kryn apko bs ak insan success kr saqta ha or wo insan ap khud hn apky elawa koi nahe ap bs koshish kryn Allah pak pe strong yakeen rakha kryn or mehnat krke sb Allah ke hawaly kr dea kryn everything is possible be brave be strong stay blessed
Hum insan hamesha moat se darty or bhagty hn or moat se bachny ki koshish krty hn jabke moat ny ana hi ana ha hum

moat se nahe bach saqty humy jahanam se bachny ki koshish krni chaheay hum jahanam se bach saqty hn
Insan ko 3 chezo se dar lagta ha Moat Risk/Dolat Ezat Shohrat Fame

Moat

Humy pta hona chaheay Moat tab ani ha jab Allah chahy us se pehly puri dunya bi ak tararf ho ke apko marna chahy to apka kuch nahe beggar saqti

Risk/Dolat

Risk Dolat sb Allah pak ke hath m ha puri dunya ak taraf b ho jay na apsy apka risk cheen saqti ha na de saqti ha Ye sirf ALLAh pak ky hath m ha

Izat Zilat shohrat Fame

Izat Zilat Sirf Allah de saqty hn puri dunya mel ke b apko zra brabr b demage nahe kr saqti Agr ALLAH

Apko izat dena chahay to puri dunya ak tararf ho ke b apko 1% b nuksan nahe pohancha saqti
Phr Dar ks bat ka ??????????????
Be brave be strong Just put your trust To ALLAH
Hamesha Confident or himat se raho kbi ksi k samne mat jhuko puri dunya mel k b apka kuch b nahe beggar saqti

ZB
MY REQUEST FROM ALL OF YOU

ZB Request from All of you My family
Mjy ap sbki help or support ki zarort ha or wo ye k hum sb Mel kr Allah ka Quran ki Urdu translation logo tak pohanchy or Quran ko samjna asaaan kryn dosro k leay sb tak Allah ka Quran pohanchy it's my compaign hum log 70+ age k ho jaty hn phr b hmy namaz tak ki translation nahe pta hoti k hum Allah pak se

Kya Dua kr rhy hn so hum youngster's ko Mel kr puri takat or energy ke sath Allah ka Quran spread krna ha
Hum puri Koshish kr rhy hn k hum sb Mel k Quran ki translation or most important topics ko maximum share kry with translation hmara maksad Quran ko spread krna ha be a part of us
m apni pocket se ye sb kr rha ho Allah ka Quran spread krny k leay apne or mene sbny Marna ha ak din to q na Allah k leay Kuch kryn apni energies apni power ko bajay negative use krny k Allah k leay invest kryn khud ko be a part of us
Agr m 23 ki age m apny sare sources use kr rha ho pocket se heavy amounts give away kr rha ho Quran spread krny k leay to ap just share to kr saqty hn itna e krdyn

Contact ZB +92 300 5427579

ZB
Agr ap mujsy contact na kr sky too b ap jo b hn jaha bi hn waha Allah ka quran pheelay zada se zada with urdu translation jitni himat ha utna share kryn

MY Family

ust Relax and focus! Exams are not difficult. Put in your 100% effort and trust in Allah.

AL-Quran

"And indeed, with hardship comes ease." (Quran 94:5)
Remember, I'm here to support you! Stay brave and strong!"

May Allah bless you